

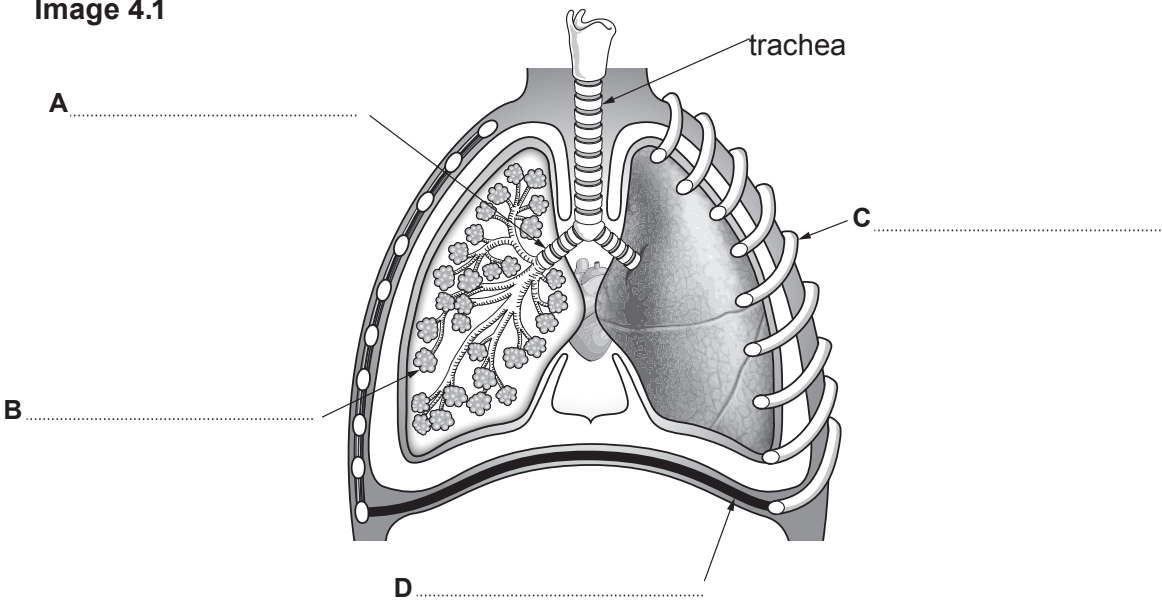
4. (a) **Image 4.1** shows a diagram of the thorax.

Choose words from the list to label parts **A** to **D** on **Image 4.1**.

[4]

bronchiole rib alveolus bronchus diaphragm membrane

Image 4.1



(b) **Images 4.2A** and **4.2B** show part of the lining of the trachea for a non-smoker and a smoker.

Lining of trachea	
Image 4.2A – non smoker	Image 4.2B – smoker
The diagram shows the tracheal lining for a non-smoker. It features a layer of cells with cilia (working cilia) that move mucus upwards, as indicated by a blue arrow. Bacteria are trapped in the mucus. Labels include: cells lining trachea, direction of movement of mucus, bacteria, mucus, and working cilia.	The diagram shows the tracheal lining for a smoker. It features a thick layer of mucus that traps bacteria. Labels include: bacteria and mucus.



Examiner
only

(i) Use **Image 4.2A** to describe how the mucus and cilia work together to remove bacteria from the trachea in a non-smoker.

I. mucus [1]

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II. cilia [1]

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(ii) Use **Image 4.2B**, to explain how smoking can lead to bacteria infecting the lungs. [2]

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(c) Smoking cigarettes can cause lung disease.

(i) State the name of **one** lung disease caused by smoking cigarettes. [1]

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(ii) In 2007, the Welsh Government banned smoking in enclosed public spaces. Suggest **one** reason for the ban. [1]

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